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Kawamoto

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(54) **VEHICLE CABIN ILLUMINATION DEVICE**

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B60Q 3/74 (2017.01)
B60Q 3/76 (2017.01)

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CPC ***B60Q 3/64*** (2017.02); *B60Q 3/74* (2017.02); *B60Q 3/76* (2017.02)

(58) **Field of Classification Search**
CPC B60Q 3/50–64; B60Q 3/70–78
See application file for complete search history.

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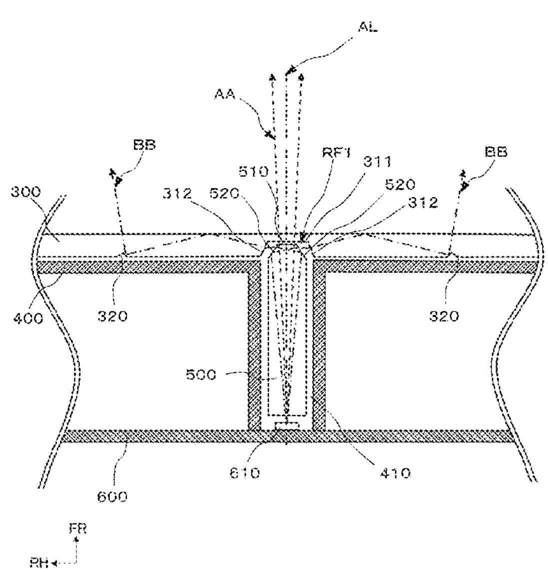
(57) **ABSTRACT**

Provided is a vehicle cabin illumination device which can be inexpensively constituted with less attenuation of light intensity. The vehicle cabin illumination device is provided with a light source, and a translucent cover, wherein the cover is provided with a light-entering portion to which illumination light from the light source is irradiated, and a reflector which is formed on a surface opposite to cabin side of a vehicle and reflects the illumination light entering the cover toward the cabin side of the vehicle, wherein the light-entering portion has a translucent portion that passes some of the illumination light from the light source toward the cabin side of the vehicle, and an incident portion that enters some of the illumination light from the light source toward inside of the cover.

7 Claims, 13 Drawing Sheets



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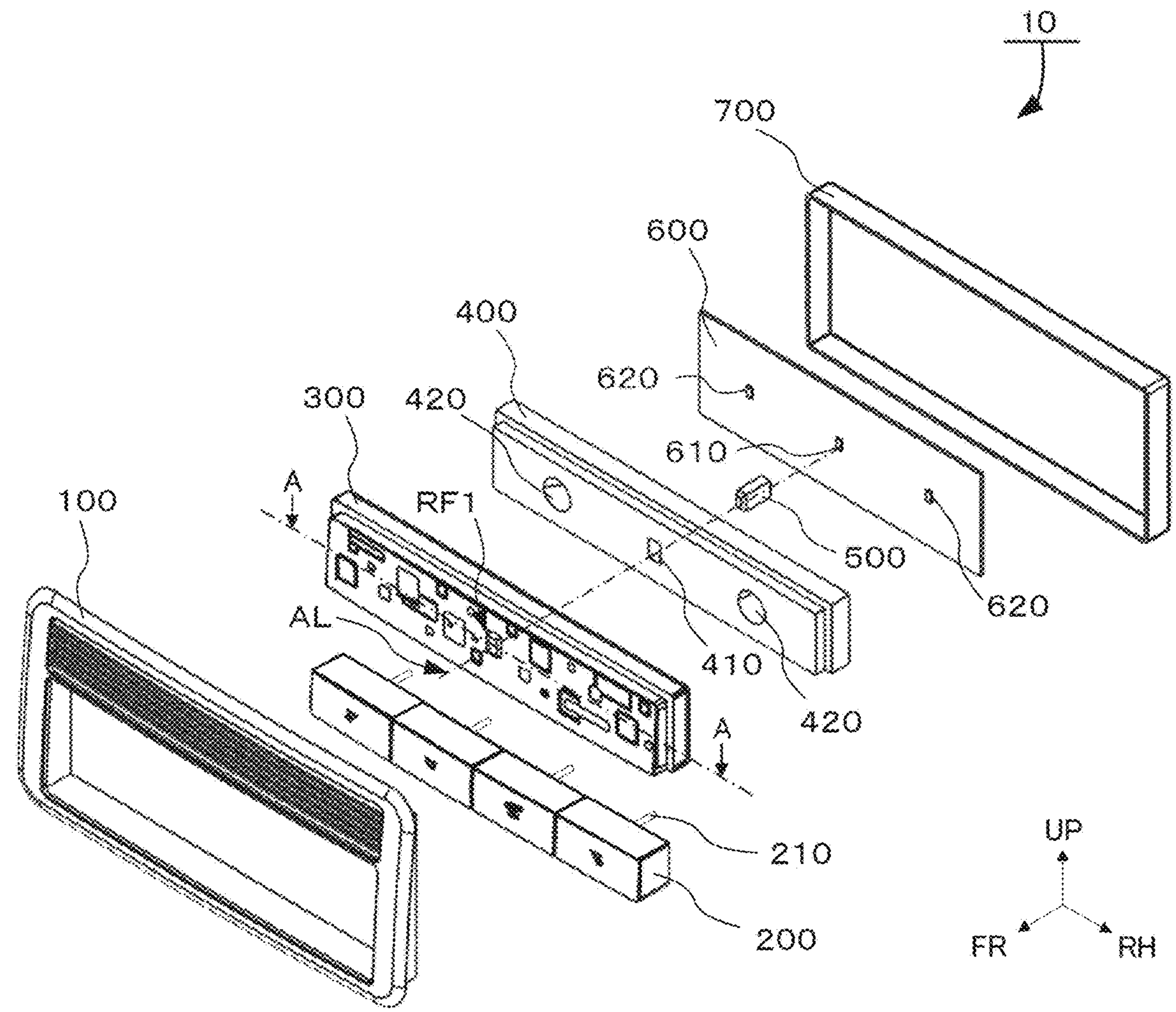

10
700
600
400
620
300
420
610
A
RF1
100
500
620
AL
410
420
A
210
UP
200
FR
RH

Fig. 1

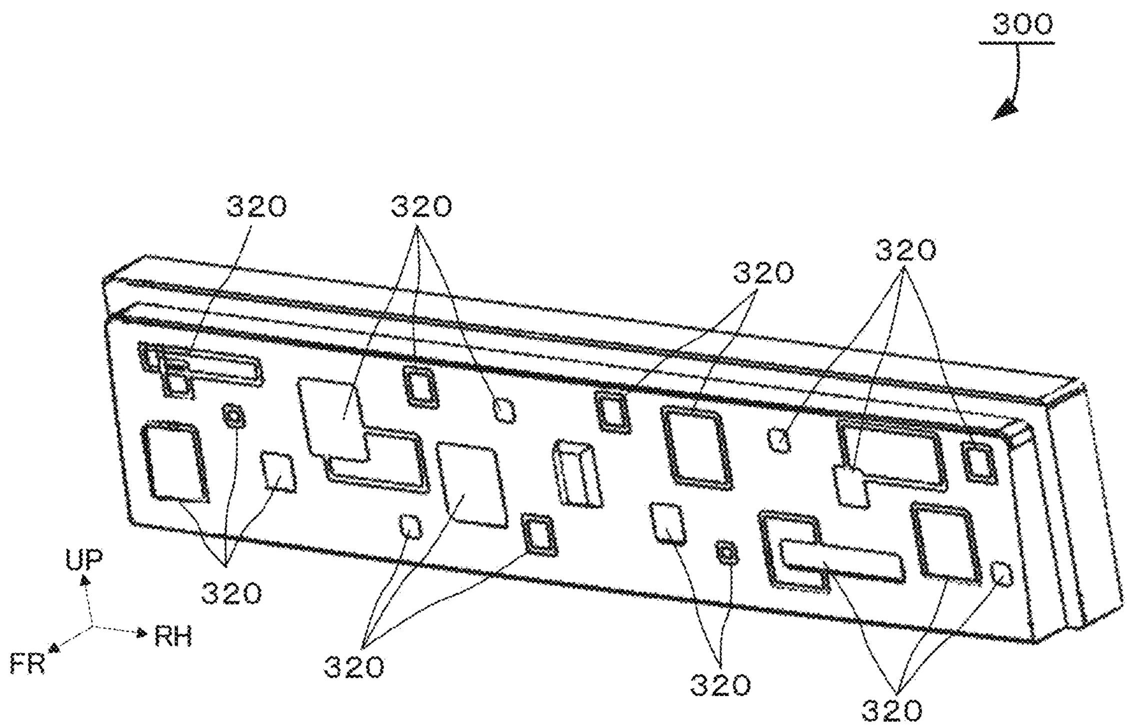


Fig. 2A

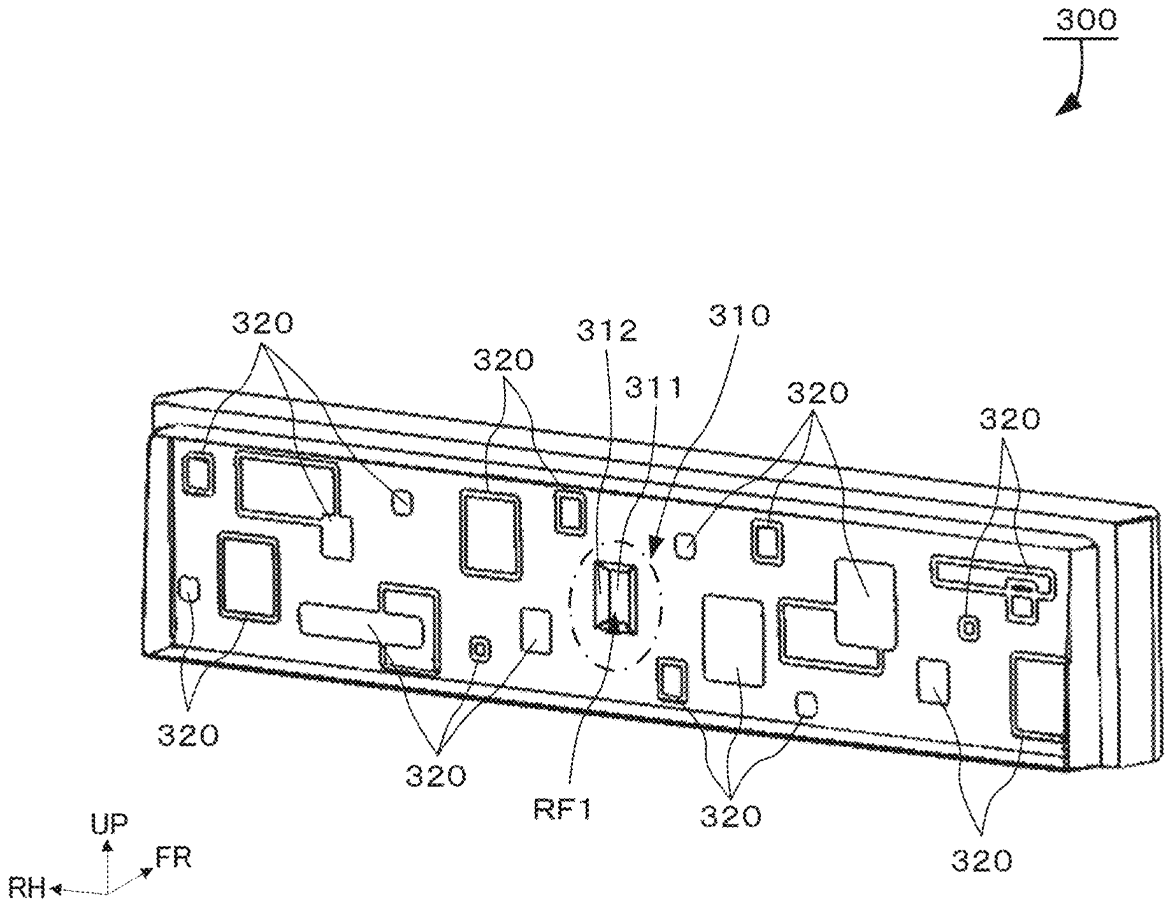
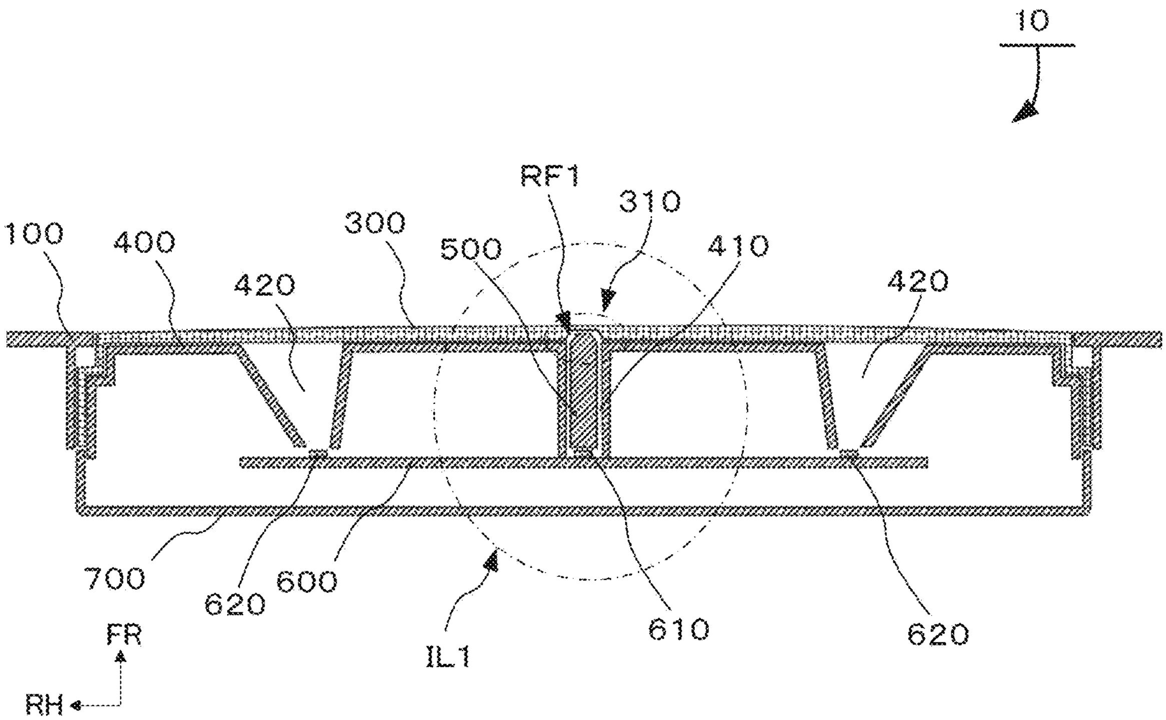


Fig. 2B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RF1
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100
400
300
500
410
420
420
700
620
600
610
620
IL1
FR
RH

Fig. 3

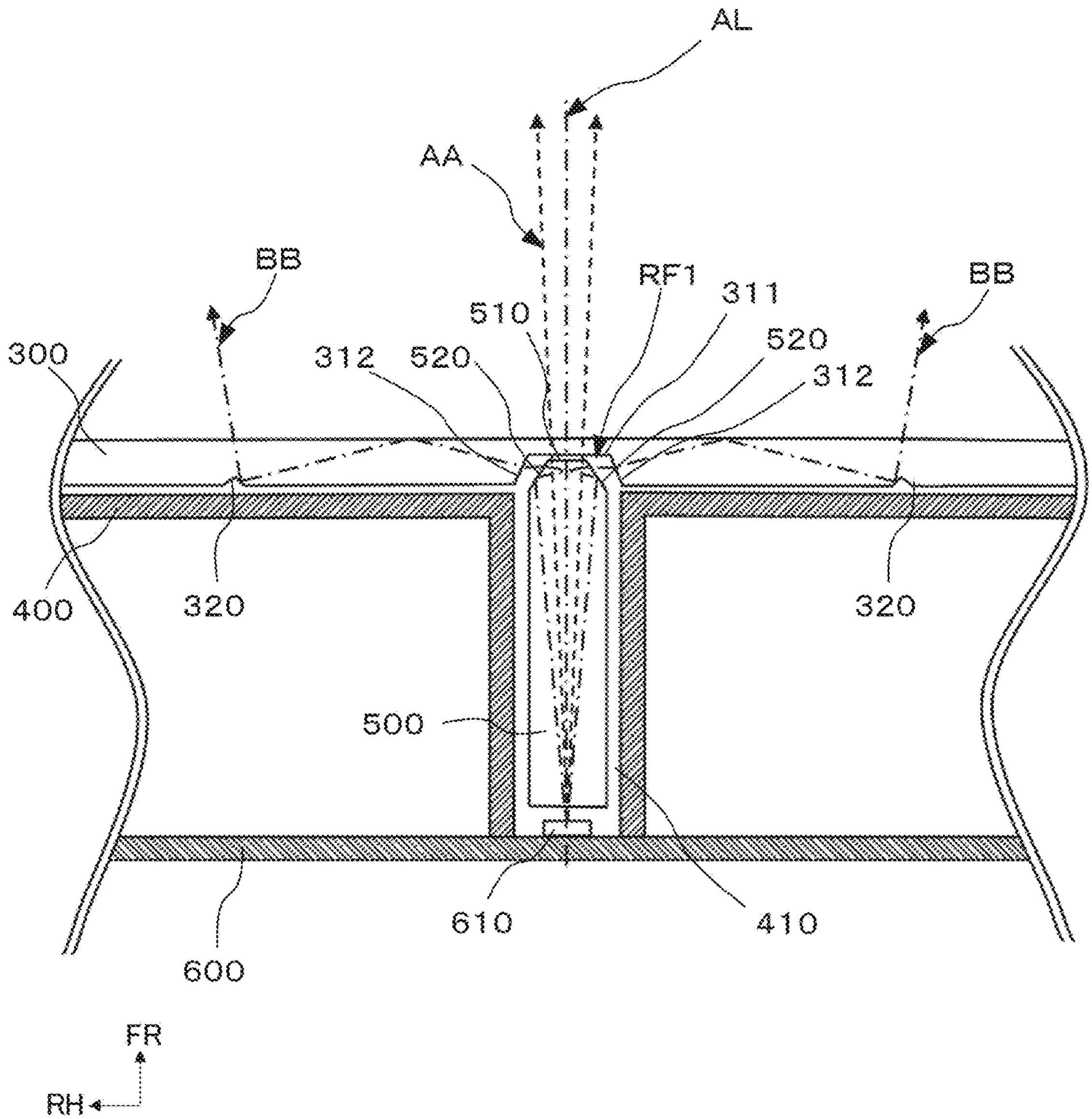

AL
AA
BB
RF1
311
510
BB
520
312
520
312
300
400
320
320
500
610
410
600
FR
RH

Fig. 4

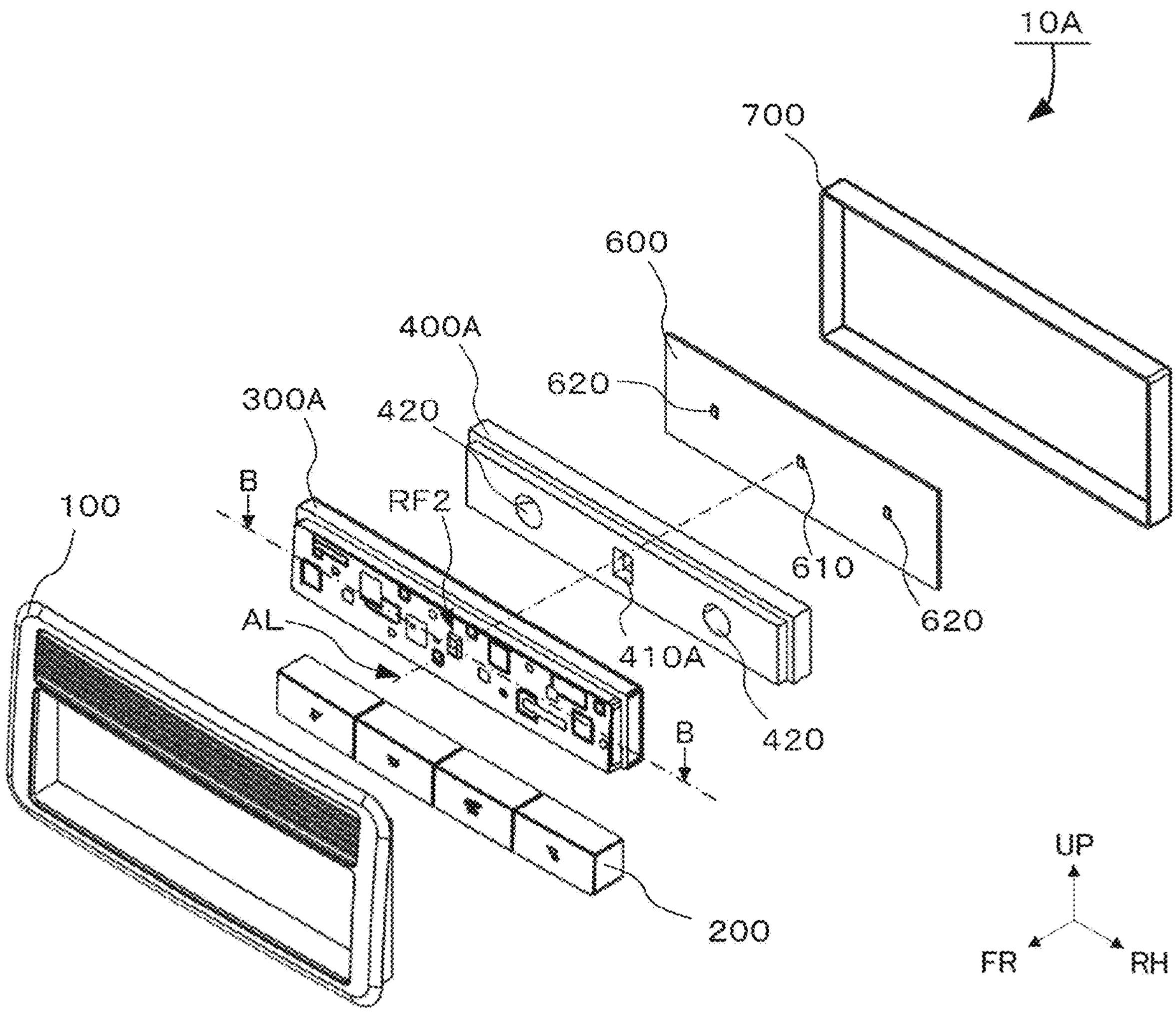
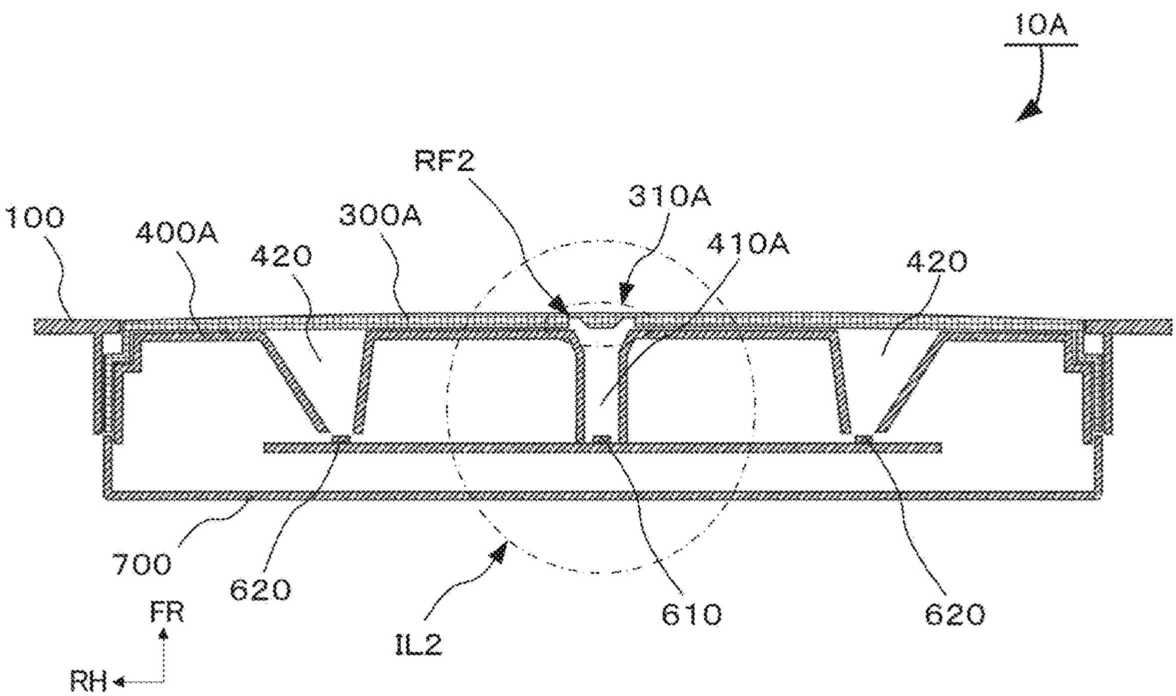

10A
700
600
400A
620
300A
420
RF2
B
100
610
620
AL
410A
420
B
200
UP
FR
RH

Fig. 5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RF2
310A
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400A
300A
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420
420
700
620
610
620
FR
RH
IL2

Fig. 6

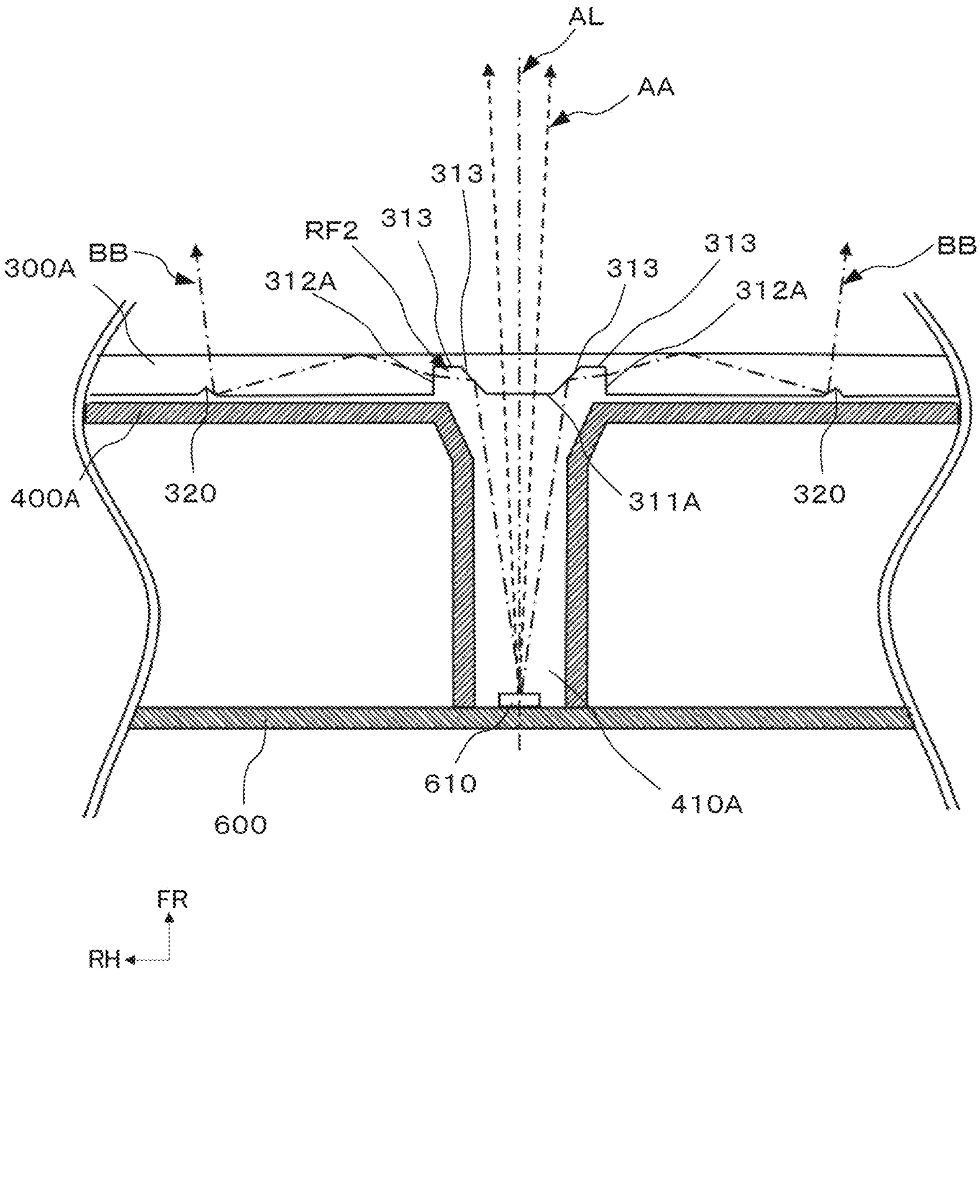
AL
AA
313
313
RF2
313
313
313
BB
300A
312A
312A
BB
400A
320
311A
320
610
410A
600
FR
RH

Fig. 7

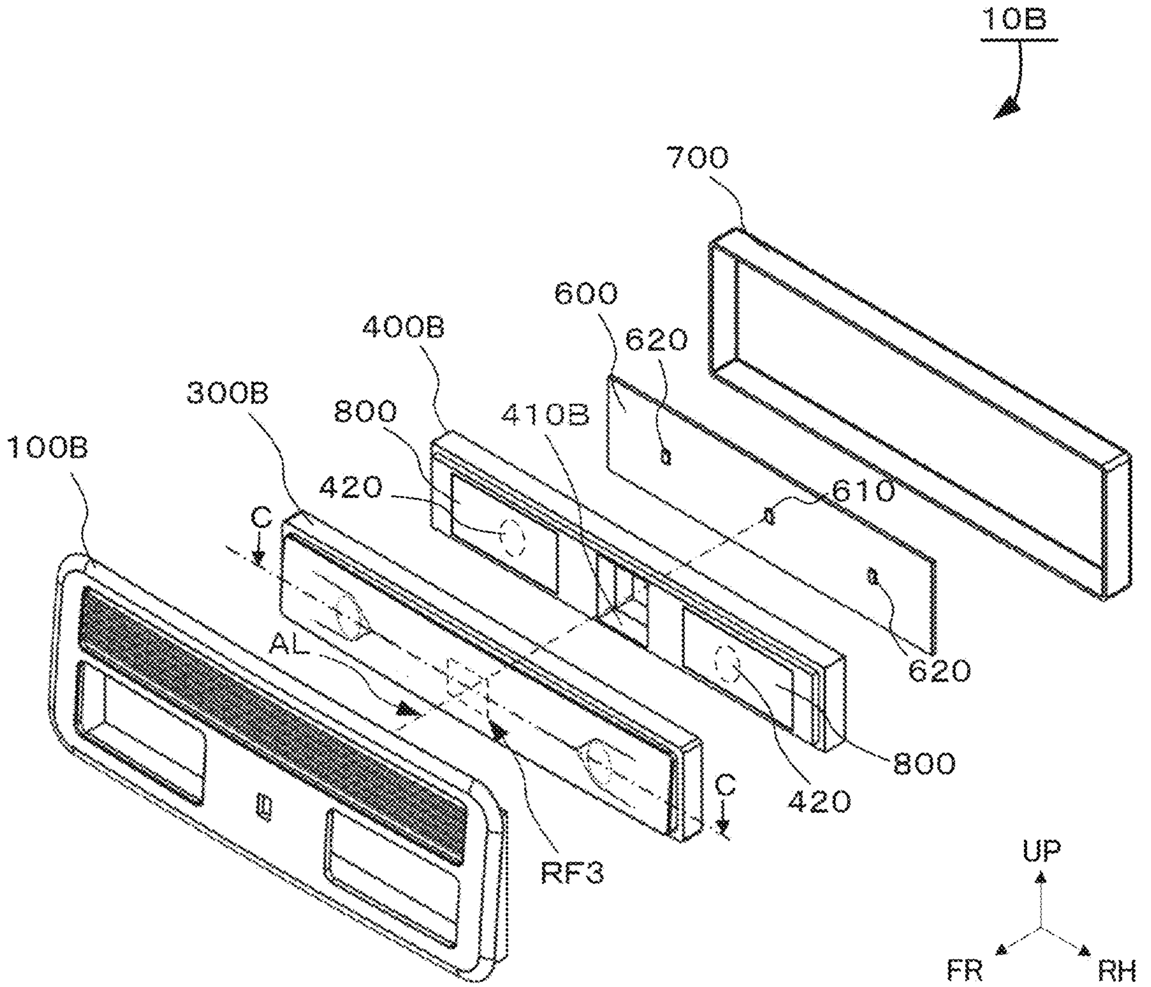
10B
700
600
620
400B
410B
300B
800
100B
420
610
C
620
AL
800
420
C
RF3
UP
FR
RH

Fig. 8

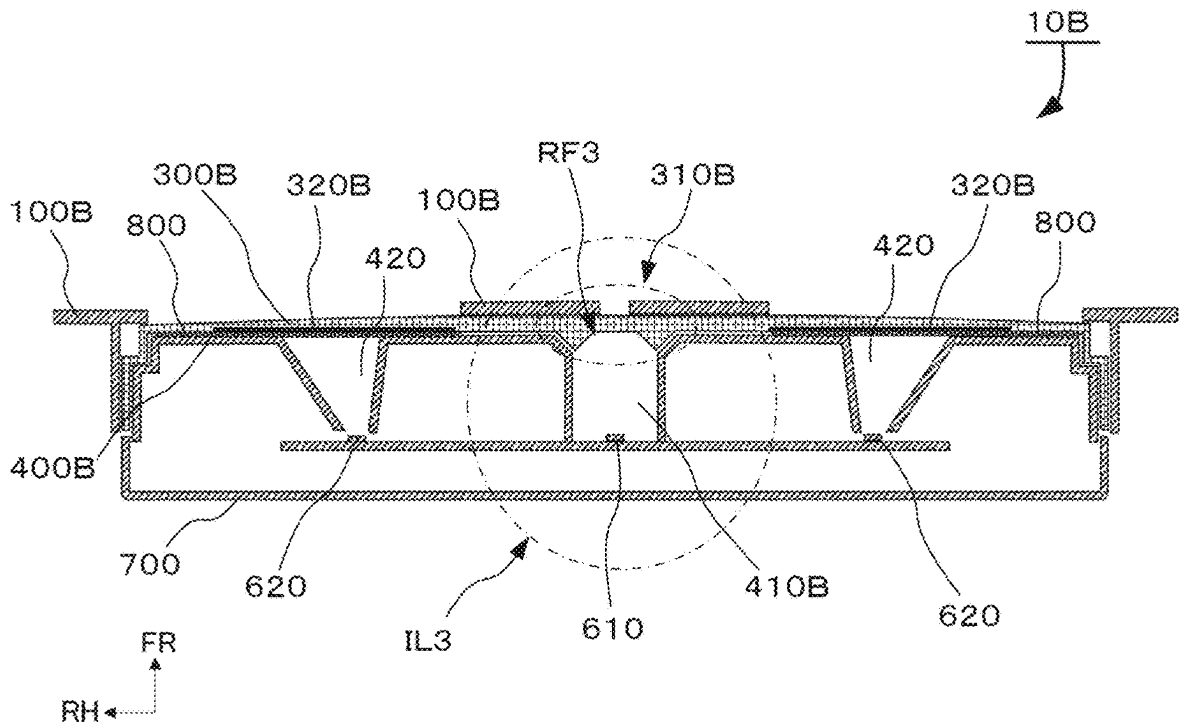
10B
RF3
300B
320B
100B
310B
320B
100B
800
420
420
800
400B
700
620
IL3
610
410B
620
FR
RH

Fig. 9

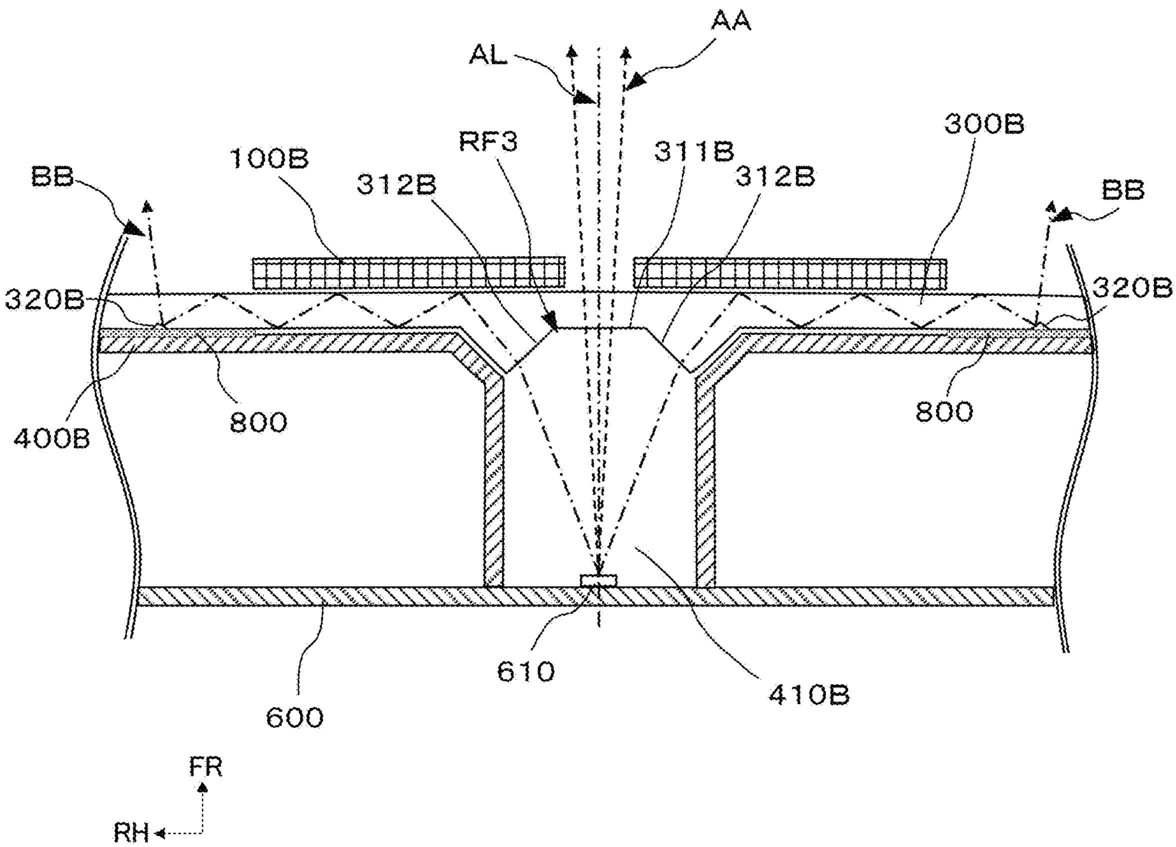
AL
AA
RF3
311B
300B
100B
312B
312B
BB
BB
320B
320B
800
800
400B
610
410B
600
FR
RH

Fig. 10

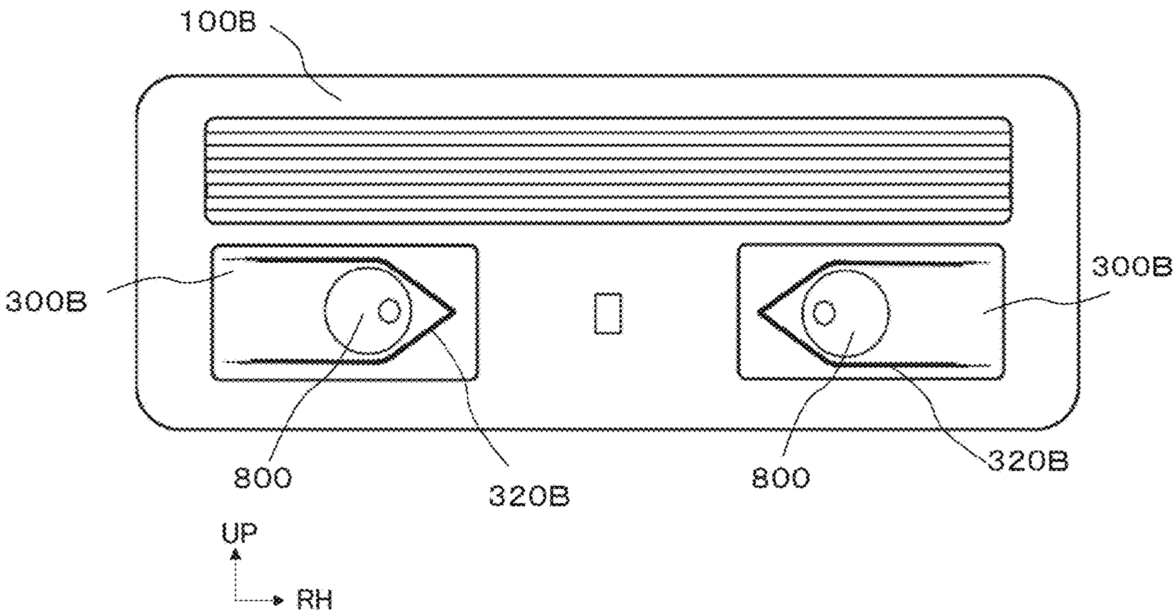

Fig. 11A

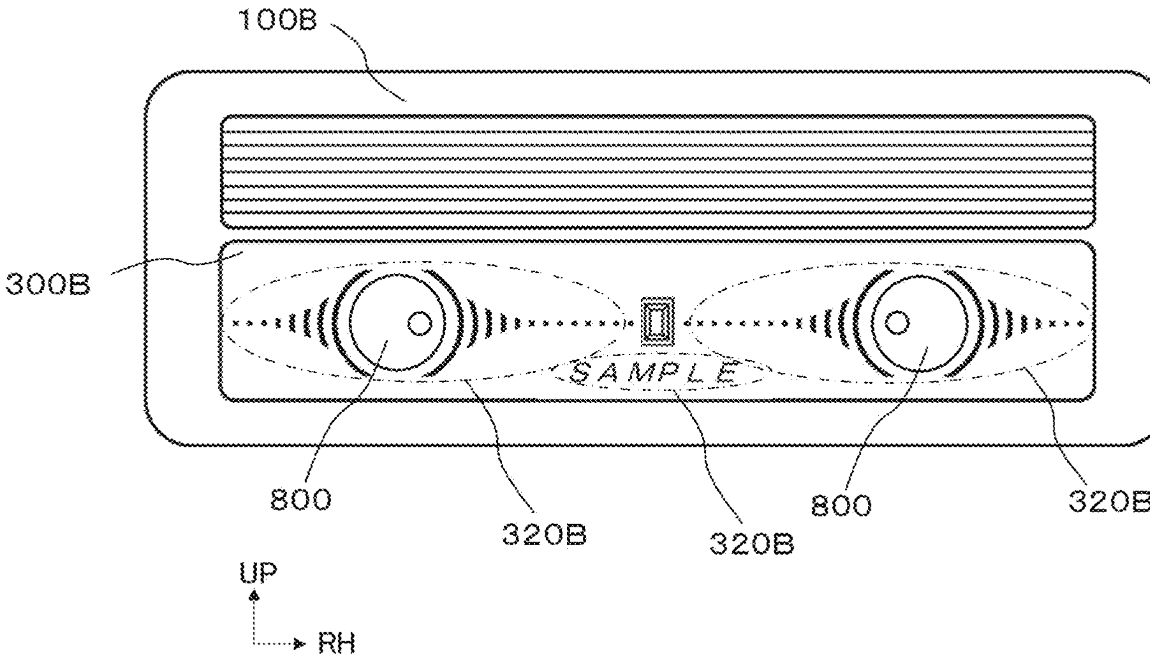

Fig.11B

VEHICLE CABIN ILLUMINATION DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a national stage entry of PCT/JP2022/45779, filed on Dec. 13, 2022, that claims the benefit of priority to Japanese Application No. 2022-057696, filed Mar. 30, 2022, the entire disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a vehicle cabin illumination device.

BACKGROUND ART

In general, a illumination device for illuminating a cabin of a vehicle comprises direct illumination for illuminating a predetermined part of the cabin and indirect illumination for improving a cabin atmosphere.

Disclosed is an illumination device of this kind, which comprises: an LED serving as a light source; a light diffusing member for diffusing light from the LED; an optical sheet subjected to hologram processing and so on for reflecting diffused light from the light diffusing member toward the cabin of the vehicle; and a translucent transparent cover having a first light transmitting portion (direct illuminating portion) for directly emitting the diffused light traveling toward the cabin of the vehicle into the cabin of the vehicle, and a second light transmitting portion (indirect illuminating portion) for emitting light reflected by the optical sheet into the cabin of the vehicle, wherein the transparent cover covers the LED, the light diffusing member, and the optical sheet from the cabin of the vehicle (For example, see Patent Literature 1).

PATENT LITERATURE

Japanese Unexamined Patent Application Publication No. 2009-214677

SUMMARY OF INVENTION

Technical Problem

However, in the technique disclosed in Patent Literature 1 above, in addition to the light diffusing member for diffusing light from the light source, it is necessary to provide the optical sheet for diffusing the diffused light for indirect illumination, and the transparent cover.

Further, there is a problem that the light emitted from the light source may pass through the transparent cover after passing through the diffused light member, whereby luminous intensity of illumination light emitted into the cabin of the vehicle may be attenuated.

Considering the above fact, the present invention aims to provide the vehicle cabin illumination device which can be configured at a low cost with less attenuation of light intensity.

Solution to Problem

Form 1

At least one embodiment of the present invention proposes a vehicle cabin illumination device comprising a light source, and a translucent a cover disposed on cabin side of a vehicle and covering the light source, wherein the cover is provided with a light-entering portion to which illumination light from the light source is irradiated, and a reflector formed on a surface opposite to the cabin side of the vehicle and reflecting the illumination light incident in the cover toward the cabin side of the vehicle, wherein the light-entering portion has a translucent portion that allows some of the illumination light from the light source to pass through to the cabin side of the vehicle, and an incident portion that allows some of the illumination light from the light source to enter toward inside of the cover.

Advantageous Effect of Invention

According to at least one embodiment of the present invention, it is possible to provide the vehicle cabin illumination device that can be configured at a low cost with less attenuation of light intensity.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a diagrammatic view of an overall configuration of a vehicle cabin illumination device according to a first embodiment.

FIG. **2** illustrates a cover illustrated in FIG. **1**, FIG. **2**A from front upward view and FIG. **2**B from the rear upward view.

FIG. **3** is a cross-sectional view (cross-sectional view at A-A line position of FIG. **1**) of the first embodiment of the vehicle cabin illumination device illustrated in FIG. **1**.

FIG. **4** is an enlarged cross-sectional view of IL**1** portion illustrated in FIG. **3** (cross-sectional view at the A-A line position in FIG. **1**).

FIG. **5** is a diagrammatic view of an overall configuration of a vehicle cabin illumination device according to a second embodiment.

FIG. **6** is a cross-sectional view (cross-sectional view at B-B line position in FIG. **5**) of the second embodiment of the vehicle cabin illumination device illustrated in FIG. **5**.

FIG. **7** is an enlarged cross-sectional view of IL**2** portion illustrated in FIG. **6** (cross-sectional view at the B-B line position of FIG. **5**).

FIG. **8** is a diagrammatic view of an overall configuration of a vehicle cabin illumination device according to a third embodiment.

FIG. **9** is a cross-sectional view (cross-sectional view at C-C line position in FIG. **8**) of the third embodiment of the vehicle cabin illumination device illustrated in FIG. **8**.

FIG. **10** is an enlarged cross-sectional view of IL**3** portion illustrated in FIG. **8** (cross-sectional view at the C-C line position in FIG. **8**).

FIG. **11** is a front view illustrating display examples FIG. **11**A and FIG. **11**B of the vehicle cabin illumination device according to the third embodiment.

DESCRIPTION OF EMBODIMENT

First Embodiment

A vehicle cabin illumination device **10** according to the present embodiment will be described using FIG. **1** to FIG. **4**. The vehicle cabin illumination device **10** of the present embodiment is an illumination device disposed on a ceiling and so on in a vehicle cabin. Incidentally, arrow FR appropriately illustrated in drawings indicates front (front view direction) of the vehicle cabin illumination device **10** illustrated in FIG. **1**, arrow UP indicates upper side, and arrow RH indicates right side. In the following description, when directions up-down, front-back, and left-right are used, vertical, front and back, and left and right directions of the vehicle cabin illumination device **10** are illustrated unless otherwise noted.

Configuration of Vehicle Cabin Illumination Device **10**

The vehicle cabin illumination device **10** according to the present embodiment is fixed to a ceiling portion and so on in a vehicle cabin as, for example, a room lamp or a reading lamp. The vehicle cabin illumination device **10** receives power supply and so on when a harness (not illustrated) is connected from a vehicle.

As illustrated in FIG. **1**, the vehicle cabin illumination device **10** according to the present embodiment includes a bezel **100**, a switch knob **200**, a cover **300**, a case **400**, a light guide lens **500**, a circuit board **600**, a light source **610**, a light source **620**, and a lid **700**.

Bezel **100**

The bezel **100** is a frame installed on cabin side of the vehicle. The bezel **100** is formed of an impermeable resin and so on in a substantially rectangular shape in front view. The bezel **100** has an opening portion in a center portion, and a switch knob **200** and a cover **300** are fitted into the opening portion. The bezel **100** is constituted as a light shielding wall for preventing illumination light from leaking out from a part other than the opening portion.

Switch Knob **200**

The switch knob **200** is an input device for performing operations such as lighting of illumination. The switch knob **200** is integrally formed of resin and so on into the substantially rectangular shape in front view. In the present embodiment, the switch knob **200** is provided with, for example, 4 kinds of functional switches, and when any of the functional switches is pressed, each of them performs independent functional operations. A substantially cylindrical rod push pin **210** extending rearward is formed behind the switch knob **200**, and a push switch (not illustrated) mounted on the circuit board **600** and a tip portion of the push pin **210** are disposed close to each other. On front side of the switch knob **200**, a plurality of symbol marks (4 parts in the present embodiment) indicating an operation position are provided.

Cover **300**

The cover **300** irradiates the cabin of the vehicle with the illumination light from the light source **610** as direct illumination (spot illumination) and indirect illumination by illuminating the entire cover **300**. The cover **300** also protects the light sources **610** and **620** by preventing contact with outside.

The cover **300** is fitted into the bezel **100** so that a front surface of the cover **300** is disposed on the cabin side of the vehicle and a rear surface is disposed so as to cover the light sources **610** and **620**. The cover **300** is in a bottomed box shape opened rearward and is formed of a permeable resin and so on. As illustrated in FIG. **2**A, the front surface of the cover **300** faces the cabin side of the vehicle and has a flat surface without any irregularities. On the other hand, the rear surface of the cover **300** has a light-entering portion **310** illustrated by a dashed line in FIG. **2**B and a reflector **320** illustrated by a broken line, and a recess portion RF**1** is formed in the light-entering portion **310**. The front surface of the cover **300** may be a curved surface.

The light-entering portion **310** is irradiated with irradiation light from the light source **610**. The light-entering portion **310** is formed opposite to the front of the light source **610** so that a center of the light-entering portion **310** is positioned on an optical axis AL from the light source **610**. The light-entering portion **310** has a recessed shape recessed forward in the center portion of the rear surface of the cover **300**. The light-entering portion **310** forms the recess portion RF**1** having a translucent portion **311** as a bottom surface and an incident portion **312** as a peripheral wall surface.

The translucent portion **311** passes some of the illumination light from the light source **610** to the cabin side of the vehicle. The illumination light from the translucent portion **311** is irradiated as the direct illumination (spot illumination). The translucent portion **311** is positioned with the optical axis AL from the light source **610** as the center, and is formed in front of the light source **610** as a substantially rectangular plane parallel to the circuit board **600**. The incident portion **312** is adjacently formed around the translucent portion **311**. Thickness of the translucent portion **311** in the front-back direction is smaller than the thickness of a part of the cover **300** on which the reflector **320** described further below is formed.

The incident portion **312** enters some of the illumination light from the light source **610** toward the inside of the cover **300**. The incident portion **312** is the substantially rectangular plane formed so as to surround the light-transmitting portion **311**, and a plurality of incident portions **312** are provided around the translucent portion **311**. The incident portion **312** is provided with an inclination in direction closer to the circuit board **600** from the center of the optical axis AL of the light source **610** toward radial outside of the optical axis AL.

The reflector **320** is formed on a surface of the cover **300** opposite to the cabin side of the vehicle. The reflector **320** reflects the illumination light entering the cover **300** toward the cabin side of the vehicle, and is formed on the rear surface of the cover **300** as an aggregate of minute projections having an inclined shape such as a substantially triangular pyramid shape. Specifically, the reflector **320** is formed by aggregating projections including minute inclined shapes in a substantially rectangular or rectangular frame shape illustrated in a range surrounded by a thick broken line in FIG. **2**B. The reflector **320** diffusely reflects the irradiated irradiation light forward.

Case **400**

As illustrated in FIG. **3**, the case **400** is disposed between the cover **300** and the circuit board **600**, and is a light guide path for guiding the irradiation light from the light sources **610** and **620** mounted on the circuit board **600** to the cover **300**. The case **400** has the bottomed box shape opened rearward, and is formed of the impermeable resin and so on. The light guide path is formed so as to surround the light source **610** and the light source **620**. The light guide path is disposed close to the circuit board **600**. The case **400** has a light guide cylinder **410** as the light guide path provided in front of the light source **610**, and a light guide cylinder **420** as a light guide path for the light source **620** provided on both sides in the left-right direction of the light guide cylinder **410**.

The light guide cylinder **410** guides the illumination light from the light source **610** to the light-entering portion **310**. The light guide cylinder **410** has a substantially rectangular opening portion in front view and is formed as a substantially rectangular penetrating-space extending in the front-back direction. The light guide cylinder **420** guides the illumination light from the light source **620** to the cabin side of the vehicle. The light guide cylinder **420** has a substantially circular opening portion in front view, and is formed as a substantially conical penetrating space extending in the front-back direction so that a diameter of a circle increases as it moves forward.

Light Guide Lens **500**

The light guide lens **500** is disposed in the light guide cylinder **410** of the case **400**, and guides the illumination light irradiated from the light source **610** to the light-entering portion **310**. An end portion of the light guide lens **500** enters into the recess portion RF**1** provided in the cover **300**.

The light guide lens **500** is a rod-shaped rectangular parallelepiped extending in the front-back direction in front view and is formed of a translucent resin and so on. A rear end portion of the light guide lens **500** is formed with the rectangular plane parallel to the circuit board **600** and opposed to the light source **610**. On the other hand, a plane opposed to the recess portion RF**1** of the cover **300** is formed at the front end of the light guide lens **500**. Specifically, a transmitting surface **510**, which is the plane parallel to the translucent portion **311**, is formed at the tip portion opposite to the translucent portion **311**. A reflecting surface **520**, which is the plane having an inclination in direction closer to the circuit board side toward the outer periphery from the side of the plane of the tip portion, is formed on a side surface of the tip portion opposite to the incident portion **312**. The transmitting surface **510** and the reflecting surface **520**, which are the end portions of the light guide lens **500**, are disposed in a state of entering into the recess portion RF**1** of the cover **300**.

Circuit Board **600**

The circuit board **600** mounts the light source **610** and the light source **620**. The circuit board **600** is formed in a substantially rectangular plate shape in front view, and is disposed so that its long sides extend in the left-right direction. The circuit board **600** is disposed in parallel with the cover **300**. The light source **610** is mounted substantially in the center of the circuit board **600**, and the light source **620** is mounted on the circuit board **600** in the left-right direction. A harness (not illustrated) is connected to the circuit board **600** in order to receive the power supply from the vehicle. The push switch (not illustrated) that receives the operations of the switch knob **200** is mounted on the circuit board **600**. The push switch is mounted opposite and close to the end portion of the push pin **210** extending from the switch knob **200**.

Lid **700**

The lid **700** is positioned at the rearmost position of the vehicle cabin illumination device **10**, and is fitted into the bezel **100** so as to cover the switch knob **200**, the cover **300**, the case **400**, the light guide lens **500**, and the circuit board **600** and forms the outer shape. The lid **700** fixes the circuit board **600** with a set screw and so on.

Effects

When the switch knob **200** for lighting is pressed, the vehicle cabin illumination device **10** configured as described above uses the power supply from the harness connected from the vehicle to light the light source **610** and the light source **620** mounted on the circuit board **600**.

As illustrated in FIG. **3**, the illumination light irradiated from the light sources **620** provided on left and right portions of the circuit board **600** passes through the cover **300** through the light guide cylinder **420** provided on the case **400**, and is emitted in the cabin of the vehicle as the direct illumination. At this time, strong illumination light is emitted to a predetermined range without being reflected and diffused until it is emitted in the cabin of the vehicle.

Further, as illustrated by a broken line arrow AA in FIG. **4**, the illumination light irradiated from the light source **610** provided in the center portion of the circuit board **600** passes through the inside of the light guide lens **500** housed in the light guide cylinder **410** and reaches the transmitting surface **510** and the reflecting surface **520** provided on the side surface of the tip portion of the light guide lens. The illumination light having reached the transmitting surface **510** is transmitted through the translucent portion **311** of the cover **300**, and is emitted in the cabin of the vehicle as the direct illumination. In this case, the strong illumination light is emitted to a narrow range without being reflected and diffused until it is emitted in the cabin of the vehicle.

On the other hand, the illumination light that has reached the reflecting surface **520** is reflected by the inclination of the reflecting surface **520** and guided towards the inside of the cover **300**, as illustrated by a dashed line arrow BB in FIG. **4**. The illumination light reflected by the reflecting surface **520** passes through the reflecting surface **520** provided on the opposite side of the reflected reflecting surface **520**, and reaches the inside of the cover **300** through the incident portion **312** provided on the light-entering portion of the cover **300**. At this time, since the reflecting surface **520** is formed with the plane having the inclination opposite to the incident portion **312**, the irradiation light reaches the inside of the cover **300** without attenuation.

The irradiation light having reached the inside of the cover **300** reaches the reflector **320** formed on the surface of the cover **300** opposite to the cabin side of the vehicle. The reflector **320** is formed as the aggregate of minute projections having the inclined shape such as the substantially triangular pyramid shape. The irradiation light irradiated on the reflector **320** is diffusely reflected forward by a minute plane formed on the reflector **320**. Further, since the reflector **320** is formed in a wide range of the cover **300**, the irradiation light is emitted from the entire plane of the cabin of the vehicle on the cover **300**. The irradiation light is emitted as the indirect illumination from the entire surface of the cover **300** facing the cabin of the vehicle.

As described above, the vehicle cabin illumination device **10** according to the present embodiment is provided with the light source **610** and the light source **620** provided on the circuit board, and the translucent cover **300** disposed on the cabin side of the vehicle and covering the light source **610** and the light source **620**. The cover **300** is provided with the light-entering portion **310** to which the illumination light from the light source **610** is irradiated, and the reflector **320** formed on the surface opposite to the cabin side of the vehicle and reflecting the illumination light entered into the cover **300** toward the cabin side of the vehicle. The light-entering portion **310** has the translucent portion **311** for allowing some of the illumination light from the light source **610** to pass toward the cabin side of the vehicle, and the incident portion **312** for allowing some of the illumination light from the light source **610** to enter the inside of the cover **300**. Therefore, the cover **300** protects the light sources **610** and **620** by covering them so that they are not in contact with the outside, and splits the irradiation light from the light source **610** by the translucent portion **311** and the incident portion **312**, whereby the direct illumination and the indirect illumination can be irradiated from one light source **610**. The light-entering portion **310** is provided with the recess portion RF**1** having the bottom surface serving as the translucent portion **311** and a side wall serving as the incident portion **312**. The incident portion **312** is provided with the inclination in the direction closer to the circuit board **600** from the center of the optical axis of the light source **610** toward the radial outside of the optical axis. Therefore, by providing the incident portion **312** with the inclination on the incident portion **310** to which the irradiation light from the light source **610** is irradiated, the irradiation light used for the indirect illumination can be spectrally separated with a simple configuration and less attenuation. The light guide cylinder **410** as the light guide path from the light source **610** to the light-entering portion **310** is provided between the cover **300** and the circuit board **600**. The translucent portion **311** and the incident portion **312** of the recess portion RF**1** are disposed so as to oppose to the internal space of the light guide cylinder **410** and the optical axis of the light source **610**. The light guide lens **500** for guiding the light from the light source **620** to the light-entering portion **310** is disposed in the light guide cylinder **410** as the light guide path. The end portion of the light guide lens **500** enters the recess portion RF**1**. Therefore, the irradiation light irradiated from the light source **610** is guided to the light-entering portion **310** in the state surrounded by the light guide cylinder **410**. The transmitting surface **510** and the reflecting surface **520** provided at the end portion of the light guide lens **500** enter the recess portion RF**1** in which the light-entering portion **310** is provided, so that the irradiation light can reach the light-entering portion **310** with less attenuation of the irradiation light. Further, since the illumination light incident from the incident portion **312** is guided to the reflector **320** through the inside of the cover **300**, the cover **300** alone can constitute a functional component of the light guiding, reflecting, and a protective cover, and can be made thin. Therefore, it is possible to have the simple configuration with a small number of components, and it is possible to provide an inexpensive vehicle cabin illumination device with less attenuation of light intensity.

Second Embodiment

Next, a vehicle cabin illumination device **10**A according to the present embodiment will be described using FIG. **5** to FIG. **7**. In FIG. **5** to FIG. **7**, the same reference numerals are assigned to the same components as those of a vehicle cabin illumination device **10** of a first embodiment.

Configuration of Vehicle cabin illumination device **10**A

As illustrated in FIG. **5**, the vehicle cabin illumination device **10**A according to the present embodiment includes a bezel **100**, a switch knob **200**, a cover **300**A, a case **400**A, a circuit board **600**, a light source **610**, a light source **620**, and a lid **700**.

Cover **300**A

The cover **300**A has a bottomed box shape opened rearward and is formed of a permeable resin and so on. As illustrated in FIG. **6** and FIG. **7**, the cover **300**A has a light-entering portion **310**A and a reflector **320** on a rear surface, and a recess portion RF**2** recessed forward is formed in the light-entering portion **310**A.

The light-entering portion **310**A is irradiated with irradiation light from a light source **610**. The light-entering portion **310**A is formed opposite to the front of the light source **610** so that center of the light-entering portion **310**A is positioned on an optical axis AL from the light source **610**. The light input portion **310**A has a recessed shape recessed forward in a center portion of the rear surface of the cover **300**A, and a recessed portion RF**2** is formed on radial outside of the optical axis AL with a translucent portion **311**A as a bottom surface and an incident portion **312**A as a surrounding wall surface. In the recess portion RF**2**, a convex portion having a substantially trapezoidal cross section is formed in the center portion of the bottom surface of the recess portion RF**2** by protruding a substantially rectangular plane toward the circuit board **600**. The translucent portion **311**A, the incident portion **312**A, and a reflecting surface **313** are formed in the recess portion RF**2**.

The translucent portion **311**A is the substantially rectangular plane formed on a top surface of the convex portion in the recess portion RF**2**. A plane formed on the top surface of the convex portion is formed as the plane which positions the optical axis AL from the light source **610** at the center of the top surface of the convex portion and is parallel to the circuit board **600** in front of the light source **610**.

The incident portion **312**A is the substantially rectangular plane formed by a wall surface portion outside in the radial direction of the optical axis AL of the recess portion RF**2**, and a plurality of incident portions **312**A are provided to surround the recess portion RF**2**. The incident portion **312**A is formed substantially parallel to the optical axis AL.

The reflecting surface **313** is formed by the wall surface of the convex portion at the center portion of the recess portion RF**2** and the substantially rectangular plane formed on the bottom surface of the recess portion RF**2**. A plurality of wall surfaces of the center convex portion of the recess portion RF**2** are formed so as to surround the convex portion. The wall surface of the convex portion of the center portion of the recess portion RF**2** is provided with an inclination in direction closer to a cabin of a vehicle from the center of the optical axis AL of the light source **610** toward the radial outside of the optical axis AL. The substantially rectangular plane formed on the bottom surface of the recess portion RF**2** is formed so as to surround the convex portion and is formed in front of the light source **610** as the plane parallel to the circuit board **600**. The reflecting surface **313** is formed in a mirror-like shape by vapor deposition and so on.

Case **400**A

The case **400**A is disposed between the cover **300**A and the circuit board **600**, and is a light guide path for guiding the irradiation light from the light sources **610** and **620** mounted on the circuit board **600** to the cover **300**A.

As illustrated in FIG. **7**, the light guide cylinder **410**A formed in the case **400**A guides illumination light from the light source **610** to the light-entering portion **310**A. The light guide cylinder **410**A has a substantially rectangular opening portion in front view, and is formed as a substantially rectangular penetrating space extending in front-back direction. Further, a cabin end portion of the vehicle of the light guide cylinder **410**A is inclined and widened toward the radial outside of the optical axis AL so that the illuminating light from the light source **610** is irradiated to the translucent portion **311**A of the outside.

Effects

When the switch knob **200** for lighting is pressed, the vehicle cabin illumination device **10**A configured as described above lights the light source **610** and the light source **620** mounted on the circuit board **600** by using power supply from a harness connected from the vehicle.

As illustrated by a broken line arrow AA in FIG. **7**, the illumination light irradiated from the light source **610** provided at the center portion of the circuit board **600** passes through inside of the light guide cylinder **410**A, passes through the translucent portion **311**A of the cover **300**A, and is emitted in the cabin of the vehicle as direct illumination. In this case, strong illumination light is emitted to a narrow range without being reflected and diffused until it is emitted in the cabin of the vehicle.

On the other hand, the illumination light that has reached the reflecting surface **313** is reflected by the inclination of the reflecting surface **313**, as illustrated by a dashed line arrow BB of in FIG. **7**, and reaches the inside of the cover **300**A through the incident portion **312**A. At this time, the reflecting surface **313** is formed with the plane having the inclination opposite to the incident portion **312**A on which the illumination light is incident, so that the illumination light reaches the inside of the cover **300**A without attenuation.

The irradiation light having reached the inside of the cover **300**A reaches the reflector **320** formed on a surface of the cover **300**A on the side opposite to the cabin side of the vehicle, and diffusely reflects forward. The illumination light is emitted as indirect illumination from the entire surface of the cover **300**A facing the cabin of the vehicle.

As described above, in the vehicle cabin illumination device **10**A according to the present embodiment, the light-entering portion **310**A of the cover **300**A comprises: the recess portion RF**2** having a side wall serving as the incident portion **312**A; the translucent portion **311**A protruding from the bottom surface of the recess portion RF**2** toward the circuit board **600** and its top surface is opposed to the light source **610**; and a reflecting surface **313** having the outer peripheral surface of the translucent portion **311**A inclined in the radial direction and reflecting light from the light source **610** toward the incident portion **312**A. That is, the translucent portion **311**A, the incident portion **312**A, and the reflecting surface **313** are provided on the light-entering portion **310**A to which the irradiation light from the light source **610** is irradiated. By irradiating one light source **610** to one light incoming portion **310**A, the irradiation light can be distributed to the irradiation light passing through the translucent portion **311**A used for the direct illumination and the irradiation light passing through the incident portion **312**A used for the indirect illumination. In addition, since the illumination light incident from the incident portion **312**A is guided to the reflector **320** through the inside of the cover **300**A, the cover **300**A alone can constitute a functional component of the spectroscopic, light guiding, reflecting, and a protective cover, and can be made thin. Therefore, it is possible to have a simple configuration with a small number of components, and to provide an inexpensive vehicle cabin illumination device with less attenuation of light intensity.

Third Embodiment

Next, a vehicle cabin illumination device **10**B according to the present embodiment will be described using FIG. **8** to FIG. **10**. In FIG. **8** to FIG. **10**, the same reference numerals are assigned to the same components as those of a vehicle cabin illumination device **10** of a first embodiment.

Configuration of Vehicle Cabin Illumination Device **10**B

As illustrated in FIG. **8**, the vehicle cabin illumination device **10**B according to the present embodiment includes a bezel **100**B, a cover **300**B, a case **400**B, a circuit board **600**, a light source **610**, a light source **620**, a lid **700**, and an operation detector **800**.

Bezel **100**B

The bezel **100**B has opening portions on left and right sides, for example, and the cover **300**B is fitted into the opening portions. The bezel **100**B has a substantially rectangular opening portion penetrating in front-back direction for emitting irradiation light of direct illumination at a center portion. The bezel **100**B is configured as a light shielding wall for preventing illumination light from leaking out from a part other than the opening portion. The opening portion of the bezel **100**B may have a shape in which right and left opening portions are coupled.

Cover **300**B

The cover **300**B has a bottomed box shape opened rearward and is formed of a permeable resin and so on. As illustrated in FIG. **9**, the cover **300**B has a light-entering portion **310**B on a rear surface, and a recess portion RF**3** recessed forward is formed in the light-entering portion **310**B.

The light entering portion **310**B is irradiated with irradiation light from a light source **610**. The light-entering portion **310**B is formed to face the front of the light source **610** so that the center of the light-entering portion **310**B is positioned on an optical axis AL from the light source **610**. The light-entering portion **310**B has a translucent portion **311**B, which is formed in a substantially rectangular plane in front view, as a bottom surface in the center portion of the rear surface of the cover **300**, and the recess portion RF**3** formed on radial outside of the optical axis AL with an incident portion **312**B as a surrounding wall surface. In the recess portion RF**3**, the translucent portion **311**B and the incident portion **312**B are formed.

The translucent portion **311**B is formed as the substantially rectangular plane formed on the bottom surface of the recess portion RF**3**. A plane of the translucent portion **311**B is positioned with the optical axis AL from the light source **610** as a center, and is formed as the plane parallel to the circuit board **600** in front of the light source **610**.

The incident portion **312**B is the substantially rectangular plane formed on a wall surface portion on the radial outside of the optical axis AL of the recess portion RF**3**, and a plurality of incident portions **312**B are provided to surround the recess portion RF**3**. The incident portion **312**B is provided with an inclination in direction closer to the circuit board **600** from the center of the optical axis AL of the light source **610** toward the radial outside of the optical axis AL.

Reflectors **320**B are disposed in groups near the operation detector **800** to be described further below so as to have a shape indicating an operation position of the operation detector **800**. Alternatively, the reflectors **320**B are disposed in groups as to form a shape of a figure, a number or a character. The reflector **320**B displays a symbol mark or a logo indicating the operation position with illumination light by forming angles, heights, shapes and so on of respective reflectors in different shapes.

Case **400**B

The case **400**B is disposed between the cover **300**B and the circuit board **600**, and is a light guide path for guiding the irradiation light from the light sources **610** and **620** mounted on the circuit board **600** to the light-entering portion **310**B.

As illustrated in FIG. **9**, the light guide cylinder **410**B formed in the case **400**B guides the illumination light from the light source **610** to the light-entering portion **310**B. The light guide cylinder **410**B has the substantially rectangular opening portion in front view and is formed as a substantially rectangular penetrating space extending in the front-back direction.

Operation Detector **800**

The operation detector **800** is, for example, a capacitance sensor for detecting a user operation. The operation detector **800** is formed on a transparent sheet that transmits the irradiation light from rear. The operation detector **800** is disposed between the cover **300**B and the case **400**B at a position corresponding to the opening portion of the bezel **100**B, and is connected to the circuit board **600** by a flexible board and so on (not illustrated). An operation of the operation detector **800** is detected by a control circuit (not illustrated) of the circuit board **600**, and the control circuit executes lighting and so on. The operation detector **800** may be formed on the circuit board **600** instead of on the transparent sheet.

Effects

When the operation detector **800** for lighting detects the user operation, the vehicle cabin illumination device **10**B configured as described above uses power supplied from a harness connected from a vehicle to light the light source **610** and the light source **620** mounted on the circuit board **600**.

As illustrated by a broken line arrow AA in FIG. **10**, the illumination light irradiated from the light source **610** provided in the center portion of the circuit board **600** passes through inside of the light guide cylinder **410**B, passes through the translucent portion **311**B of the cover **300**B, and is emitted as the direct illumination in a cabin of the vehicle from the opening portion provided in the center portion of the bezel **100**B. In this case, strong illumination light is emitted to a narrow range without being reflected and diffused before it is emitted in the cabin of the vehicle.

On the other hand, the illumination light that has reached the incident portion **312**B reaches the inside of the cover **300**A through the incident portion **312**B without attenuation, as illustrated by a dashed line arrow BB of in FIG. **10**.

The irradiation light having reached the inside of the cover **300**B reaches the reflector **320**B formed on the surface of the cover **300**B opposite to the cabin side of the vehicle and diffusely reflects forward. As illustrated in FIG. **11**A and FIG. **11**B, the reflector **320**B is formed near the operation detector **800**, and is disposed so as to irradiate the illumination light to an intended position by forming angles, heights, shapes, and so on of the respective reflectors into different shapes. Therefore, the emitted illumination light can be shaped to indicate the operation position. Alternatively, the emitted illumination light may be in the form of the figure, the number, or the character.

As described above, the vehicle cabin illumination device **10**B according to the present embodiment comprises the operation detector **800** for detecting an operation, and the reflectors **320**B are disposed in groups so that they form the shape indicating the operation position of the operation detector **800** near the operation detector **800**. Alternatively, in the vehicle cabin illumination device **10**B according to the present embodiment, the reflectors **320**B are disposed so as to be grouped in the shapes of the figure, the number, and the character. Therefore, by guiding the illumination light emitted from one light source **610** to one light-entering portion **310**A to the reflectors **320**B formed in different shapes with different angles, heights, shapes and so on of the respective reflectors, the shape such as a symbol mark indicating the operation position or a design such as the logo can be displayed with the illumination light. In addition, since the illumination light incident from the incident portion **312**B is guided to the reflector **320** through the inside of the cover **300**A, the cover **300**B alone can constitute a functional component of the spectroscopic, light guiding, reflection, and a protective cover, and can be made thin. Therefore, it is possible to have a simple configuration with a small number of components, and it is possible to provide an inexpensive vehicle cabin illumination device with improved added value and less attenuation of light intensity.

In the first to third embodiments, the reflector **320** and the reflector **320**B are formed toward the inside (front) from the rear surface of each of the cover **300** to the cover **300**B, but may be formed to protrude outward (rear) from the rear surface of each of the cover **300** to the cover **300**B.

In the first to third embodiments, the light sources **610** and **620** are configured to be lit simultaneously, but they may be configured to be lit separately.

In the first to third embodiments, the light source **610** is lit in response to the operation of the switch knob **200**, but it may be configured to light on and off in conjunction with a headlight of the vehicle.

Although the embodiment of the present invention has been described in detail with reference to the drawings, the specific configuration is not limited to the present embodiment, and designs and so on within a range not departing from the gist of the present invention are also included.

EXPLANATION OF THE REFERENCE NUMERALS

10: Vehicle cabin illumination device
100: Bezel
100B: Bezel
200: Switch Knob
300: Cover
300A: Cover
300B: Cover
311: Translucent portion
311A: Translucent portion
311B: Translucent portion
312: Incident portion
312A: Incident portion
312B: Incident portion
313: Reflecting surface
320: Reflector
400: Case
400A: Case
400B: Case
410: Light guide cylinder
500: Light guide lens
600: Circuit board
610: Light source
700: Lid
800: Operation detector
AL: Optical axis
RF**1**: Recess portion
RF**2**: Recess portion
RF**3**: Recess portion

The invention claimed is:

1. A vehicle cabin illumination device comprising:

a light source; and

a translucent cover disposed on cabin side of a vehicle and covering the light source,

wherein the cover is provided with a light-entering portion to which illumination light from the light source is irradiated, and a reflector formed on a surface opposite to the cabin side of the vehicle and reflecting the illumination light entering the cover toward the cabin side of the vehicle,

wherein the light-entering portion has a translucent portion that allows some of the illumination light from the light source to pass through to the cabin side of the vehicle, and an incident portion that allows some of the illumination light from the light source to enter toward inside of the cover.

2. The vehicle cabin illumination device according to claim **1**, wherein the light entering portion comprises a recess portion having a bottom surface as a translucent portion and a side wall as the incident portion,

wherein the incident portion is provided with an inclination in direction closer to a circuit board from a center of an optical axis from the light source toward a radial outside of the optical axis.

3. The vehicle cabin illumination device according to claim **2** wherein a light guide cylinder serving as a light guide path from the light source to the light entering portion is provided between the cover and the light source, and the translucent portion and the incident portion of the recess portion are disposed so as to oppose to an internal space of the light guide cylinder portion and the optical axis direction.

4. The vehicle cabin illumination device according to claim **3** wherein a light guide lens for guiding light from the light source to the light entering portion is disposed in the light guide path, and an end portion of the light guide lens enters into the recess portion.

5. The vehicle cabin illumination device according to claim **1**,

wherein the light entering portion comprises:

a recess portion having a side wall as the incident portion;

the translucent portion protruding from a bottom surface of the recess portion toward a circuit board and its top surface is opposed to the light source; and

a reflecting surface formed so that an outer peripheral surface of the translucent portion is inclined in radial direction and reflects light from the light source toward the incident portion.

6. The vehicle cabin illumination device according to claim **1** comprising an operation detector for detecting an operation,

wherein the reflectors are disposed in groups near the operation detector so as to form a shape indicating an operation position of the operation detector.

7. The vehicle cabin illumination device according to claim **1** wherein the reflectors are disposed in groups to form a figure, a number, or a character shape.

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